

Real-Time Audio FFT Spectrum Analyzer

*ESP32-C3 + Basys 3 FPGA Hardware Accelerated FFT
with VGA Frequency Spectrum Display*

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Platform: ESP32-C3 Mini | Basys 3 (Xilinx Artix-7)

Tools: Arduino IDE | Xilinx Vivado 2023.2

Abstract

This project presents the design and implementation of a real-time audio frequency spectrum analyzer built from discrete hardware components. The system captures acoustic signals using a CZN-15E electret microphone, digitizes them with the internal ADC of an ESP32-C3 microcontroller, and transmits the raw time-domain samples over a high-speed SPI link to a Basys 3 FPGA development board (Xilinx Artix-7 XC7A35T). The FPGA performs a 1024-point Fast Fourier Transform (FFT) in hardware using the Xilinx FFT IP Core and renders the resulting frequency spectrum as a real-time bar graph on a 640×480 VGA monitor.

The FPGA design is structured as four cooperating Verilog modules: an SPI slave receiver with clock-domain crossing synchronizers, the Xilinx xFFT IP core configured for streaming fixed-point arithmetic, a VGA timing controller generating standard 640×480 @ 60 Hz sync signals, and a spectrum renderer that maps FFT magnitude bins to pixel columns using a logarithmic frequency axis. Additional features include a built-in sine-wave test generator, a top-3 frequency peak isolation mode, and a proximity-lockout peak sorting algorithm that prevents harmonics from being mistaken as independent peaks.

Experimental results demonstrate clean real-time visualization of tonal signals from 100 Hz to 15 kHz with correct frequency labeling. The system successfully distinguishes between ambient noise, broadband impulses (blowing into the microphone), and pure sine tones at 500 Hz and 2 kHz. The project validates the effectiveness of FPGA-accelerated DSP for real-time audio analysis at a fraction of the latency achievable in software on a single-core microcontroller.

1. Introduction

1.1 Motivation

The Fast Fourier Transform is a cornerstone algorithm in digital signal processing, enabling decomposition of time-domain signals into their constituent frequency components. While software FFT libraries exist for microcontrollers, a 1024-point FFT computed on a single-core CPU at audio data rates introduces significant latency and leaves little processing headroom for visualization tasks. By offloading the butterfly computations to an FPGA's dedicated DSP48E1 slices, throughput increases by orders of magnitude and the microcontroller is freed to focus purely on data acquisition.

This project was motivated by the desire to build a fully hardware-accelerated signal processing pipeline using affordable development boards (ESP32-C3 Mini and Basys 3), and to gain hands-on experience with SPI communication across heterogeneous hardware, Xilinx IP core integration, and real-time VGA rendering in Verilog.

1.2 Project Objectives

- Capture audio at a stable 45 kHz sample rate using the ESP32-C3 ADC.
- Stream 1024-sample frames to the FPGA over SPI at 20 MHz without data loss.
- Compute a 1024-point FFT on the FPGA using the Xilinx FFT IP Core.
- Display the real-time frequency spectrum on a 640×480 VGA monitor.
- Implement advanced display features: logarithmic frequency axis, top-3 peak isolation, and color-coded bars.
- Validate the system with known test signals (sine tones and broadband noise).

2. System Overview

2.1 System Block Diagram

The system consists of three physical hardware stages connected by standard interfaces. The complete signal path is described below:

1. **Stage 1 – Audio Capture:** The CZN-15E electret microphone outputs an analog voltage proportional to sound pressure. This voltage is fed directly into GPIO 3 of the ESP32-C3 Mini, which is configured as an ADC input with 11 dB attenuation to cover the full 0–3.1 V input range.
2. **Stage 2 – Sample Acquisition & Buffering:** The ESP32-C3 firmware collects exactly 1024 samples at a 22 μ s interval (~45 kHz). Each sample is a 12-bit value stored as a 16-bit unsigned integer in a flat audio buffer.
3. **Stage 3 – SPI Transmission:** After collecting the full frame, the ESP32 pulls the Chip Select line LOW and transfers all 2048 bytes (1024 $\times$ 16-bit samples) to the FPGA over the SPI bus running at 20 MHz.
4. **Stage 4 – FPGA SPI Reception:** The Basys 3's SPI slave receiver module reassembles the incoming bit stream into 16-bit samples, handling clock domain crossing between the 20 MHz SPI clock and the 100 MHz FPGA system clock using double-flop synchronizers.
5. **Stage 5 – FFT Computation:** The Xilinx xFFT IP Core (1024-point, fixed-point, natural order I/O) processes the incoming stream of complex samples and outputs 512 usable frequency bins in magnitude form.
6. **Stage 6 – Spectrum Rendering & VGA Output:** The spectrum renderer maps FFT magnitudes to bar heights on a 512-pixel-wide display area using a logarithmic index mapping. The VGA controller generates 640 $\times$ 480 @ 60 Hz sync signals, and the composited image is output on the 12-bit VGA port.

2.2 Interface Summary

Interface	From	To	Speed / Format
ADC	CZN-15E Mic	ESP32-C3 GPIO 3	12-bit, ~45 kHz
SPI	ESP32-C3	Basys 3 Pmod JA	20 MHz, Mode 0, MSB-first
AXI-S	SPI Receiver	xFFT IP Core	100 MHz internal bus
AXI-S	xFFT Core	Spectrum Renderer	100 MHz internal bus
VGA	Basys 3	VGA Monitor	640 $\times$ 480 @ 60 Hz, 12-bit color

3. Hardware Components

Component	Part / Model	Role in System
Microcontroller	ESP32-C3 Mini (RISC-V, 160 MHz)	ADC sampling, SPI master, USB-Serial debug
Microphone	CZN-15E Electret Module	Analog audio capture
FPGA Board	Basys 3 – Xilinx Artix-7 XC7A35T	FFT computation, VGA output
Display	VGA Monitor (640×480)	Real-time spectrum visualization
IDE / Toolchain	Arduino IDE + Xilinx Vivado 2023.2	Firmware development & FPGA synthesis

3.1 ESP32-C3 Mini

The ESP32-C3 is a RISC-V single-core microcontroller running at up to 160 MHz with 400 KB SRAM and an integrated 12-bit ADC. In this project it serves exclusively as a data acquisition and SPI master device. The ADC is configured with 11 dB attenuation to maximize the input voltage range to approximately 3.1 V, suitable for the output swing of the CZN-15E microphone module. GPIO 3 is used as the analog input, while GPIO 4 (SCK), GPIO 6 (MOSI), and GPIO 7 (CS) form the SPI bus.

3.2 CZN-15E Microphone Module

The CZN-15E is an electret condenser microphone capsule typically mounted on a small breakout board with a built-in amplifier. It outputs an analog voltage proportional to the acoustic pressure. The module is powered from 3.3 V and its analog output is connected directly to the ESP32-C3 ADC input. No external anti-aliasing filter was used in this prototype; the ADC's inherent bandwidth and sampling rate provide natural high-frequency roll-off.

3.3 Basys 3 FPGA Development Board

The Basys 3 is built around the Xilinx Artix-7 XC7A35T FPGA, which contains 33,280 logic cells, 50 DSP48E1 slices, 1,800 Kbits of block RAM, and a 100 MHz onboard oscillator. The board exposes a 12-bit VGA port and four Pmod headers for external I/O. In this project, Pmod JA is used as the SPI input from the ESP32. The center push button is mapped to the active-high system reset, and four slide switches provide user-selectable operating modes.

4. ESP32-C3 Firmware

4.1 Overview

The ESP32-C3 firmware (FFT.ino) has a simple two-phase structure that repeats continuously: collect exactly 1024 audio samples at a fixed time interval, then transmit all 2048 bytes to the FPGA over SPI. A 2 ms delay after each transmission gives the FPGA time to process the block before the next one arrives.

4.2 Key Design Decisions

- **~45 kHz (22 μ s period) chosen to cover the audible range (Nyquist = 22.5 kHz).** Sampling rate:
- **micros()-based busy-wait loop ensures sample-accurate timing without depending on FreeRTOS or timers.** Timing loop:
- **ADC_11db enables full 3.1 V input range to capture both quiet and loud sounds.** ADC attenuation:
- **20 MHz, MSB-first, Mode 0 — matches the Verilog SPI slave receiver's edge detection logic.** SPI speed:
- **Only MOSI and SCK are used; the ESP32 is pure transmitter with no read-back path.** No MISO:

4.3 Firmware Source Code

```
#include <SPI.h>

#define MIC_PIN 3
#define NUM_SAMPLES 1024
#define SAMPLE_PERIOD_US 22 // ~45 kHz sampling interval

uint16_t audio_buffer[NUM_SAMPLES];

// Custom SPI Pin Assignments
#define VSPI_CLK 4
#define VSPI_MISO -1 // Unused — transmitter only
#define VSPI_MOSI 6
#define VSPI_SS 7

SPIClass *vspi = NULL;

void setup() {
  Serial.begin(115200);
  analogSetAttenuation(ADC_11db); // Full 0-3.1V range
  pinMode(MIC_PIN, INPUT);

  vspi = new SPIClass(FSPI);
  vspi->begin(VSPI_CLK, VSPI_MISO, VSPI_MOSI, VSPI_SS);
  pinMode(VSPI_SS, OUTPUT);
  digitalWrite(VSPI_SS, HIGH); // CS idle-high

  vspi->beginTransaction(SPISettings(20000000, MSBFIRST, SPI_MODE0));
  Serial.println("ESP32-C3 Real-Time Streaming Engine Active");
}
```

```

void loop() {
    // Phase 1: Collect 1024 samples at a fixed 45 kHz rate
    unsigned long next_sample_time = micros();
    for (int i = 0; i < NUM_SAMPLES; i++) {
        audio_buffer[i] = analogRead(MIC_PIN);
        while (micros() < next_sample_time) {} // Busy-wait for next tick
        next_sample_time += SAMPLE_PERIOD_US;
    }

    // Phase 2: Transmit the full frame to the FPGA
    digitalWrite(VSPI_SS, LOW); // Assert CS - begins SPI frame
    vspi->transferBytes((uint8_t*)audio_buffer, NULL, NUM_SAMPLES * 2);
    digitalWrite(VSPI_SS, HIGH); // Deassert CS - ends SPI frame

    delay(2); // Allow FPGA to finish processing
}

```

5. FPGA Implementation (Verilog)

5.1 Module Architecture

The Basys 3 design consists of five Verilog modules instantiated in a top-level wrapper. All modules share the 100 MHz system clock.

- **top_fft_system:** Top-level module. Instantiates and connects all submodules. Handles switch multiplexing and LED diagnostics.
- **spi_slave_receiver:** Captures the SPI stream from the ESP32. Uses a 3-stage shift register for SCK edge detection and 2-stage synchronizers for CS and MOSI. Assembles 16-bit samples from two consecutive 8-bit bytes (little-endian).
- **xfft_0 (Xilinx IP):** 1024-point streaming FFT. AXI-Stream I/O. Fixed-point, 16-bit input/output width, natural order input and output. Configured for single-channel operation on an Artix-7 device.
- **vga_controller:** Generates 640×480 @ 60 Hz timing. Derives a 25 MHz pixel clock from the 100 MHz system clock using a 4-cycle divider. Outputs active-area pixel coordinates and a video_on blanking flag.
- **spectrum_renderer:** Core visualization engine. Reads FFT output magnitudes, stores them in a 512-entry BRAM, applies logarithmic index mapping, and drives the VGA color outputs with a color-coded bar graph.

5.2 SPI Slave Receiver

The SPI receiver solves the classic clock domain crossing (CDC) problem: the incoming SPI clock runs at 20 MHz from the ESP32, while the FPGA fabric runs at 100 MHz. Two-stage flip-flop synchronizers are used on CS and MOSI. The SCK line uses a three-stage synchronizer to enable rising-edge detection (transition from 01 to 10 in the two MSBs).

Incoming bits are shifted into an 8-bit shift register. When 8 bits are complete, the first byte is stored as the low byte. When the second byte arrives, both bytes are combined into a 16-bit sample and a sample_valid strobe is asserted for one system clock cycle.

5.3 Xilinx FFT IP Core Configuration

Parameter	Value
IP Version	xfft_v9.1 rev 10 (Vivado 2023.2)
FFT Length (NFFT_MAX)	1024 points (2^{10})
Architecture	Pipelined Streaming (C_ARCH=3)
Arithmetic	Fixed-point (no floating point)
Input / Output Width	16-bit real, 16-bit imaginary
Twiddle Width	16-bit
Input Order	Natural
Output Order	Natural
Scaling	Enabled
Target Device	xc7a35tcpg236-1 (Artix-7)

5.4 VGA Controller

The VGA controller generates standard 640×480 @ 60 Hz timing. The 100 MHz system clock is divided to produce a 25 MHz pixel clock pulse. Horizontal and vertical counters track

position within each frame (800 total horizontal, 525 total vertical pixels including blanking intervals). Active-low sync pulses are generated in the blanking regions.

Timing parameters: H_ACTIVE=640, H_FP=16, H_SYNC=96, H_BP=48 → H_TOTAL=800.
V_ACTIVE=480, V_FP=10, V_SYNC=2, V_BP=33 → V_TOTAL=525.

5.5 Spectrum Renderer

5.5.1 Magnitude Calculation

The Xilinx FFT outputs complex data as 16-bit real and 16-bit imaginary components packed into a 32-bit AXI-Stream word. The spectrum renderer approximates the magnitude using the alpha-max/beta-min method: $\text{magnitude} \approx \max(|\text{Re}|, |\text{Im}|) + 0.5 \times \min(|\text{Re}|, |\text{Im}|)$. This avoids the need for a square root and is accurate to within ~12% of the true magnitude. A 1-bit left shift ($\times 2$ gain boost) is applied to improve display sensitivity for quiet sounds.

5.5.2 Logarithmic Frequency Axis

Human hearing is logarithmic. Displaying FFT bins on a linear axis would compress all perceptually important low-frequency content into the leftmost few pixels. The renderer uses a piecewise linear approximation of a logarithmic mapping, dividing the 512-pixel display width into 8 segments of 64 pixels each. Each segment maps to an exponentially growing range of FFT bins, spanning from bin 2 (~88 Hz) at the left edge to bin 223+ (~10 kHz) at the right edge. Frequency axis labels at 100 Hz, 500 Hz, 1k, 2k, 5k, 10k, and 15k Hz are rendered in cyan using a 5×3 pixel bitmap font stored in a case statement.

5.5.3 Top-3 Peak Isolation Mode (sw[3])

When slide switch 3 is HIGH, the renderer activates top-3 peak mode. A streaming sorter runs in parallel with the FFT data write-back, maintaining three ranked peak slots (m1, m2, m3). A proximity lockout rule prevents two slots from being assigned to FFT bins within 4 positions of each other, ensuring the three peaks correspond to three distinct frequency components rather than adjacent bins on a single spectral peak. In this mode, only the three peak bars are rendered: peak 1 in red, peaks 2 and 3 in yellow. All other bars are suppressed to black.

5.6 Switch Controls

Switch	Name	Function
sw[0]	Test Mode	HIGH: use internal sine wave generator; LOW: use SPI microphone data
sw[1]	Test Frequency	HIGH: double the test sine frequency (~4 kHz); LOW: base frequency (~2 kHz)
sw[3]	Top-3 Peak Filter	HIGH: display only the three dominant frequency peaks with color coding

6. VGA Display & Visualization

6.1 Screen Layout

The 640×480 display is organized as follows:

- Rows 0–439: Black background area above the spectrum bars.
- Row 440: Bottom of all spectrum bars (baseline).
- Rows 0–440: Spectrum bars rendered from row 440 upward, height proportional to FFT magnitude.
- Row 441: Horizontal frequency axis line (gray).
- Rows 442–444: Tick marks at 100 Hz, 500 Hz, 1k, 2k, 5k, 10k, 15k Hz positions.
- Rows 450–454: Frequency label text (100, 500, 1k, 2k, 5k, 10k, 15k) in cyan 5×3 bitmap font.
- Columns 0–63 and 576–639: Left and right margins (black).
- Columns 64–575: Active spectrum display area (512 pixels wide).

6.2 Color Scheme

Element	Color	RGB / Hex
Normal spectrum bars	Yellow	R=F, G=F, B=0
Dominant peak (sw[3])	Red	R=F, G=0, B=0
Peaks 2 & 3 (sw[3])	Yellow	R=F, G=F, B=0
Frequency labels	Cyan	R=0, G=F, B=F
Axis line & ticks	Gray	R=7, G=7, B=0
Background	Black	R=0, G=0, B=0

6.3 Constraint File (XDC) Pin Assignments

The Basys 3 XDC constraint file maps the FPGA top-level ports to physical package pins. Key assignments are summarized below:

- **System Clock:** W5 (100 MHz oscillator)
- **Reset:** U18 (center push button, active high)
- **SPI Inputs (Pmod JA):** spi_cs→J1, spi_mosi→L2, spi_sck→G2
- **VGA Red [3:0]:** N19, J19, H19, G19
- **VGA Green [3:0]:** D17, G17, H17, J17
- **VGA Blue [3:0]:** J18, K18, L18, N18
- **H_Sync / V_Sync:** P19 / R19
- **LED [15:0]:** U16–L1 (diagnostic indicators)

7. Results & Testing

7.1 Physical Hardware Setup

The photo below shows the assembled Basys 3 board with the VGA cable and Pmod JA SPI connections to the ESP32-C3 module.

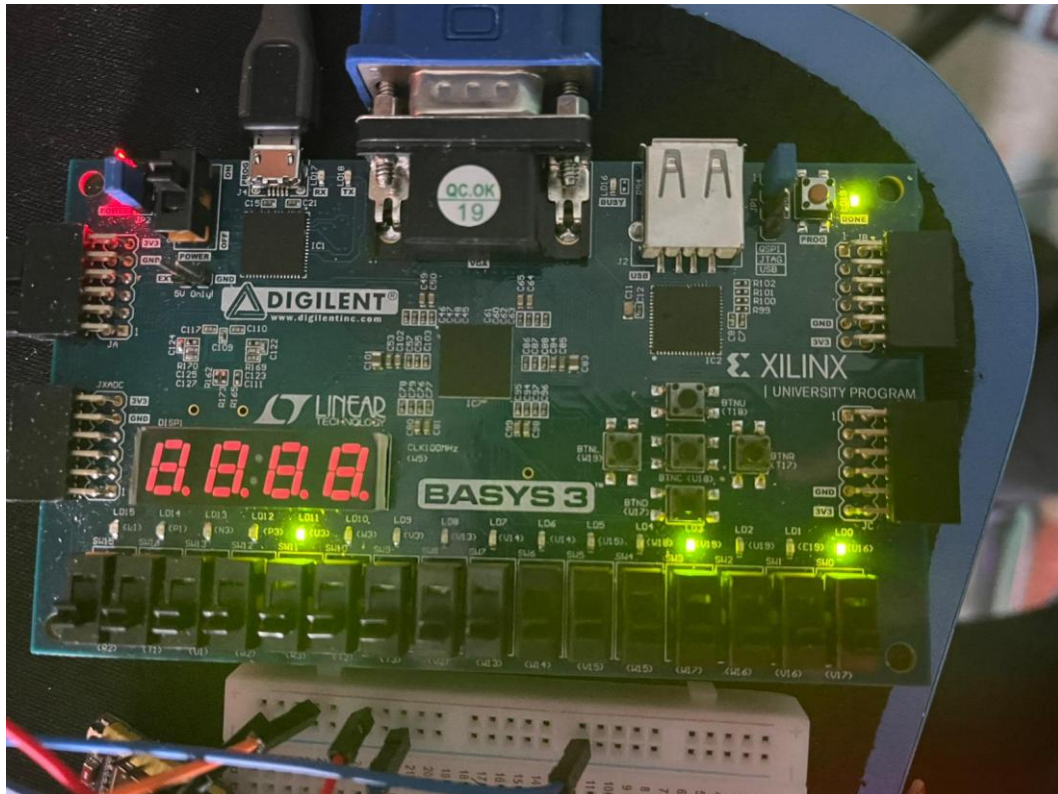


Figure 1 – Physical setup: Basys 3 FPGA board with VGA and Pmod connections

7.2 Idle Noise Floor

With no audio input and the microphone connected, the system shows a low-level noise floor dominated by ambient room noise. The bars remain short and spread across the low-frequency region. This is consistent with expected ADC quantization noise and residual 50/60 Hz hum.

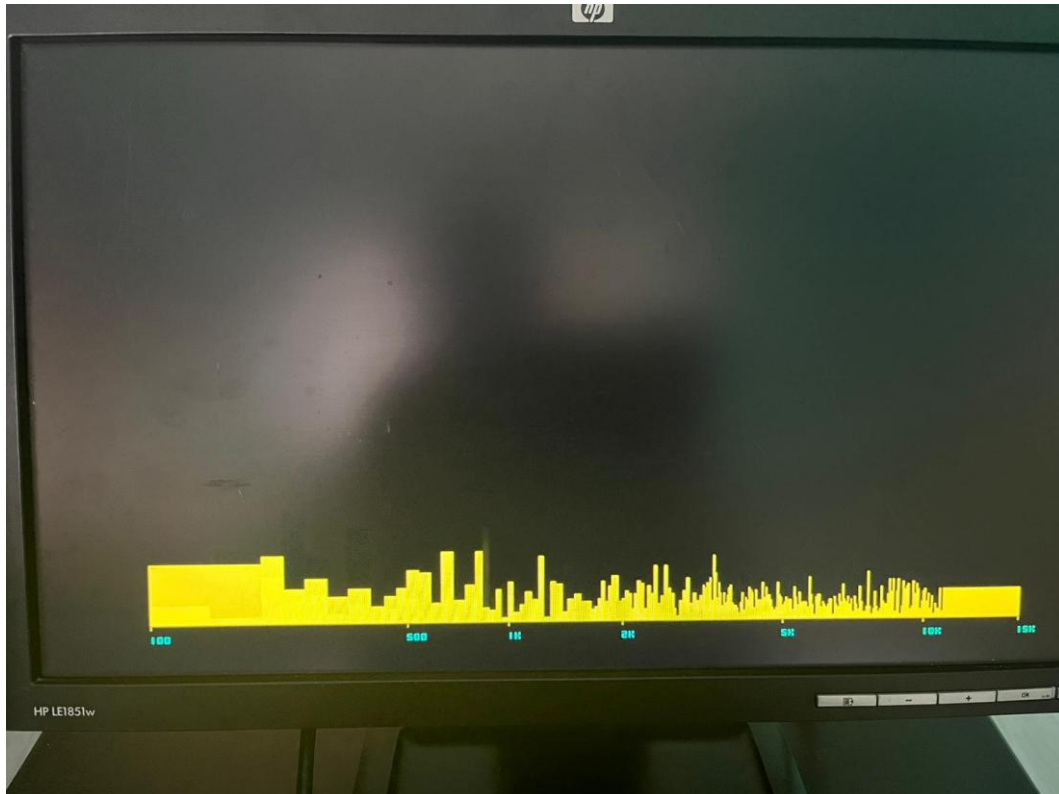


Figure 2 – Idle noise floor (no intentional audio input)

7.3 Broadband Impulse (Blowing into Mic)

Blowing directly into the microphone generates a broadband pressure impulse, which activates FFT bins across a wide frequency range. The display shows a dense cluster of bars spanning the lower and mid frequencies, visually confirming that broadband noise is correctly represented across multiple frequency bins.

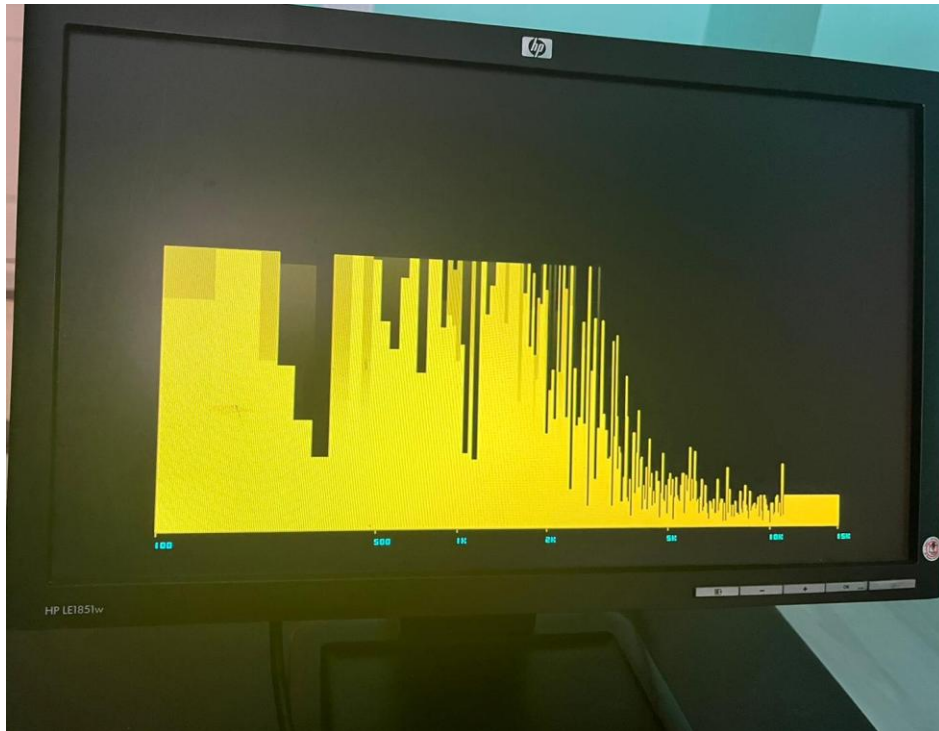


Figure 3 – VGA display when blowing into the microphone (broadband low-frequency burst)

7.4 500 Hz Sine Tone

A 500 Hz sine tone was played near the microphone. The two screenshots below show the response with the full spectrum (sw[3]=OFF) and with the Top-3 Peak isolation mode active (sw[3]=ON). In the isolation mode, the dominant 500 Hz peak is clearly highlighted in red, with two smaller harmonic peaks in yellow.



Figure 4 – 500 Hz tone, full spectrum mode (sw[3] OFF): peak visible at ~500 Hz column



Figure 5 – 500 Hz tone, Top-3 Peak mode (sw[3] ON): dominant peak in red, harmonics in yellow

7.5 2 kHz Sine Tone

A 2 kHz sine tone was applied. The FFT correctly resolves the peak at the 2 kHz position on the logarithmic frequency axis. Both screenshots confirm correct peak placement and the absence of spurious bins.

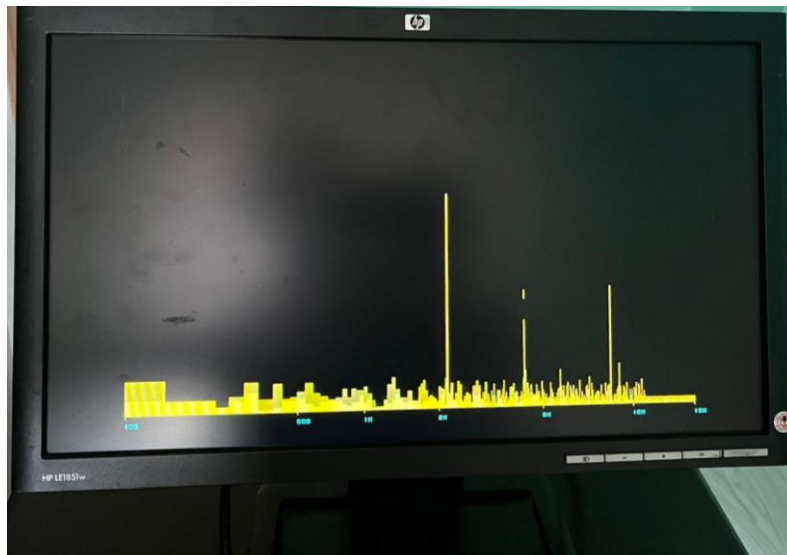


Figure 6 – 2 kHz tone, full spectrum mode



Figure 7 – 2 kHz tone, alternate view

8. Conclusion

This project successfully demonstrates a complete, end-to-end real-time audio spectrum analyzer built from off-the-shelf development hardware. The ESP32-C3 provides a stable 45 kHz audio acquisition pipeline, and the SPI link reliably delivers 1024-sample frames at 20 MHz to the FPGA without dropped data. The Basys 3 FPGA leverages the Xilinx FFT IP Core to compute hardware-accelerated 1024-point FFTs far faster than any equivalent software implementation on the same microcontroller.

The VGA visualization correctly resolves tonal signals across the 100 Hz to 15 kHz range. The logarithmic frequency axis provides a perceptually natural display, and the Top-3 Peak isolation mode with proximity-lockout sorting provides a clean, jitter-free identification of dominant frequencies even in the presence of harmonics and spectral leakage.

Potential improvements for future work include: adding a windowing function (Hann or Blackman) to reduce spectral leakage before the FFT; implementing the OLED status display on the ESP32-C3 to show the detected peak frequency numerically; upgrading the SPI link to full double-buffering using the FPGA's BRAM to allow continuous capture and display without the 2 ms gap between frames; and increasing the FFT size to 2048 or 4096 points for higher frequency resolution.

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